

Krishnaa Vadivel | Software Engineer | Scientific Computing | HPC Workflows

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Summary

Software engineer and Yale PhD researcher building scientific-computing, automation, and high-performance workflows in Python and C++. Experience spans distributed HPC, CUDA/OpenGL tooling, containerized development, numerical software, and production support around both research and engineering systems.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Develop Python/C++ workflows for first-principles and many-body simulation, using MPI, OpenMP, PETSc, SLEPc, CMake, and Linux-based automation across leadership-class HPC environments.
- Build reusable tooling around distributed compute, scientific pipelines, and ML-assisted workflows for materials and molecular problems.
- Work across research code, documentation, and theory-to-implementation translation for scalable computational workflows.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA/OpenGL software for waveform processing, interactive visualization, and diagnostic tooling for large engineering datasets.
- Developed ASP-style services and Microsoft SQL-backed workflows for internal engineering and field-data systems.
- Contributed embedded-adjacent engineering support, giving strong systems-level context for software used alongside deployed hardware.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

DOLFINx Semiconductor Workflow: Python finite-element and PETSc-based workflow for Poisson and drift-diffusion style simulation.

Agentic Scientific Automation: Retrieval-driven tooling that reads documentation and papers to generate structured computational-science workflow inputs.

Distributed Scientific Tooling: Reusable software workflows for simulation setup, engineering analysis, and container-oriented research infrastructure.

Scientific Visualization Tooling: CUDA/OpenGL and Python-based utilities for inspecting and communicating large numerical datasets.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University*MS, Electrical Engineering**2023***Cornell University***MEng, Electrical and Computer Engineering**2017–2018***Cornell University***BS, Electrical and Computer Engineering**2013–2017*

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, SQL**Scientific Computing:** MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib**Systems:** Linux, CMake, Docker, Docker Compose, Kubernetes, gcloud**Platforms:** CUDA, OpenGL, Frontier, Frontera, Perlmutter**ML / Automation:** PyTorch, scikit-learn, agentic workflows, scientific automation